ANALYZING PARIS OLYMPICS 2024 DATA

July 18, 2025

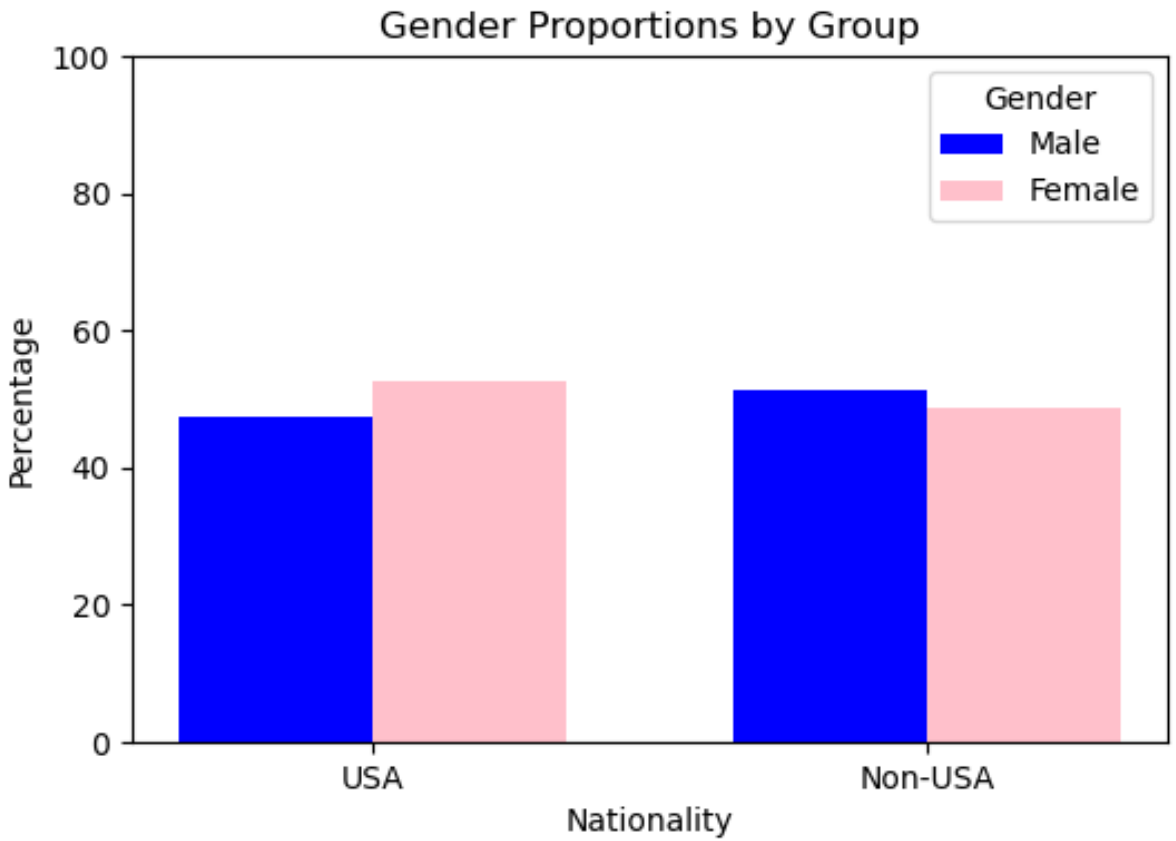
Introduction and Overview

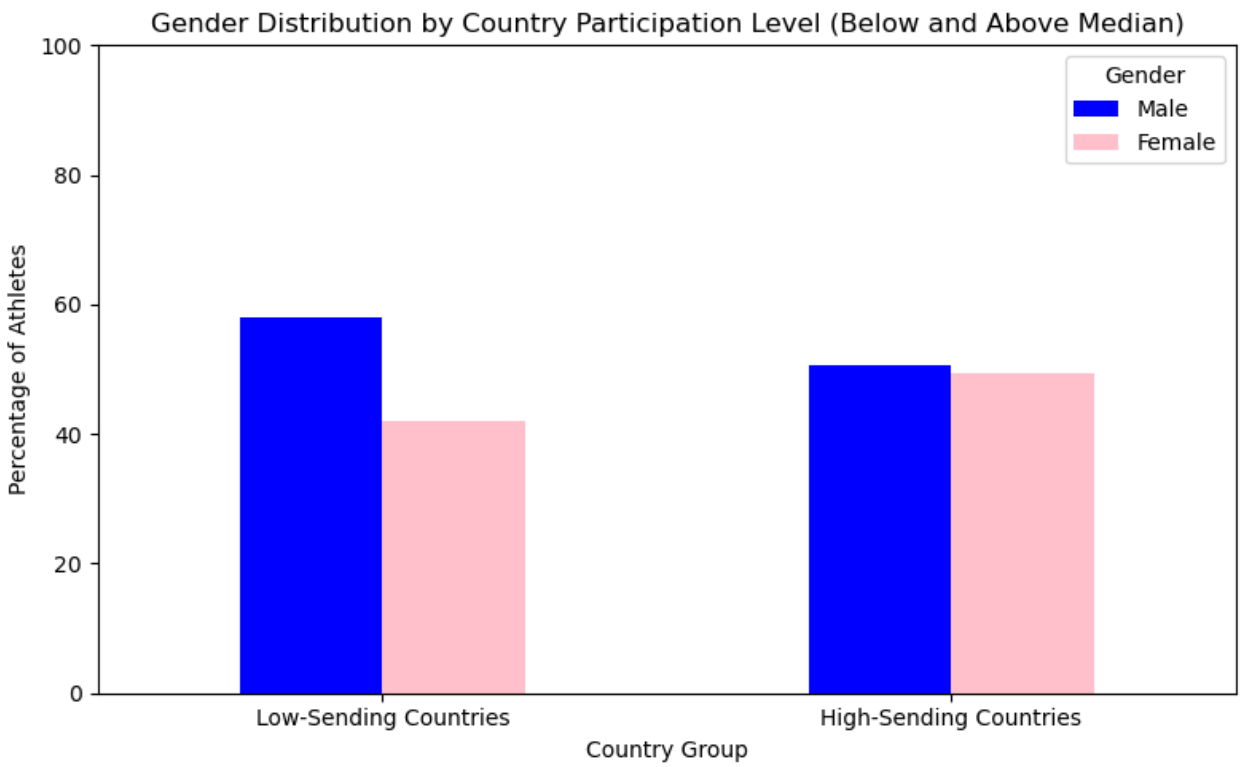
- Topics to Explore:
  - Athletes Demographics Analysis
  - Social Factors and Participation Trends
  - Performance Metrics and Statistical Analysis

- Datasets Used:
  - 2024 Paris Olympic Athlete Data
  - World Bank – GDP Data (Through 2023)

- Key data categories:
  - Athlete age, gender, height, weight, nationality, and medal status
  - Sports participation data
  - GDP information by country

Athlete Demographics: Gender



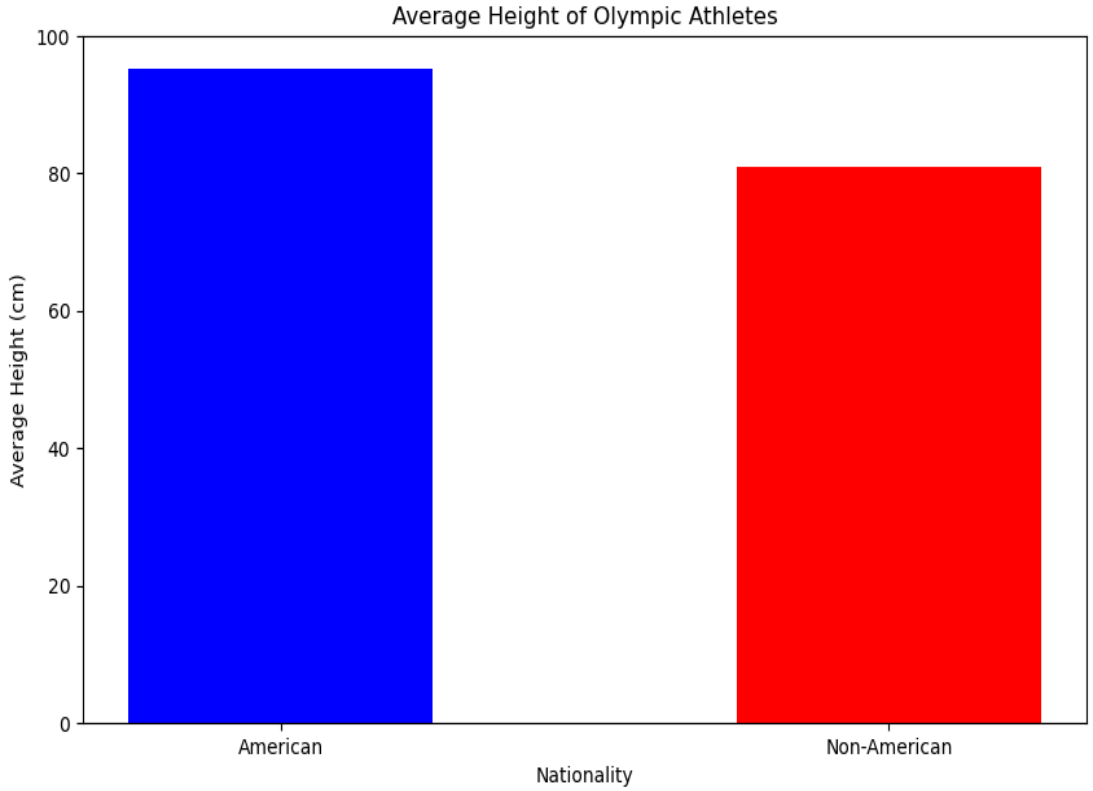


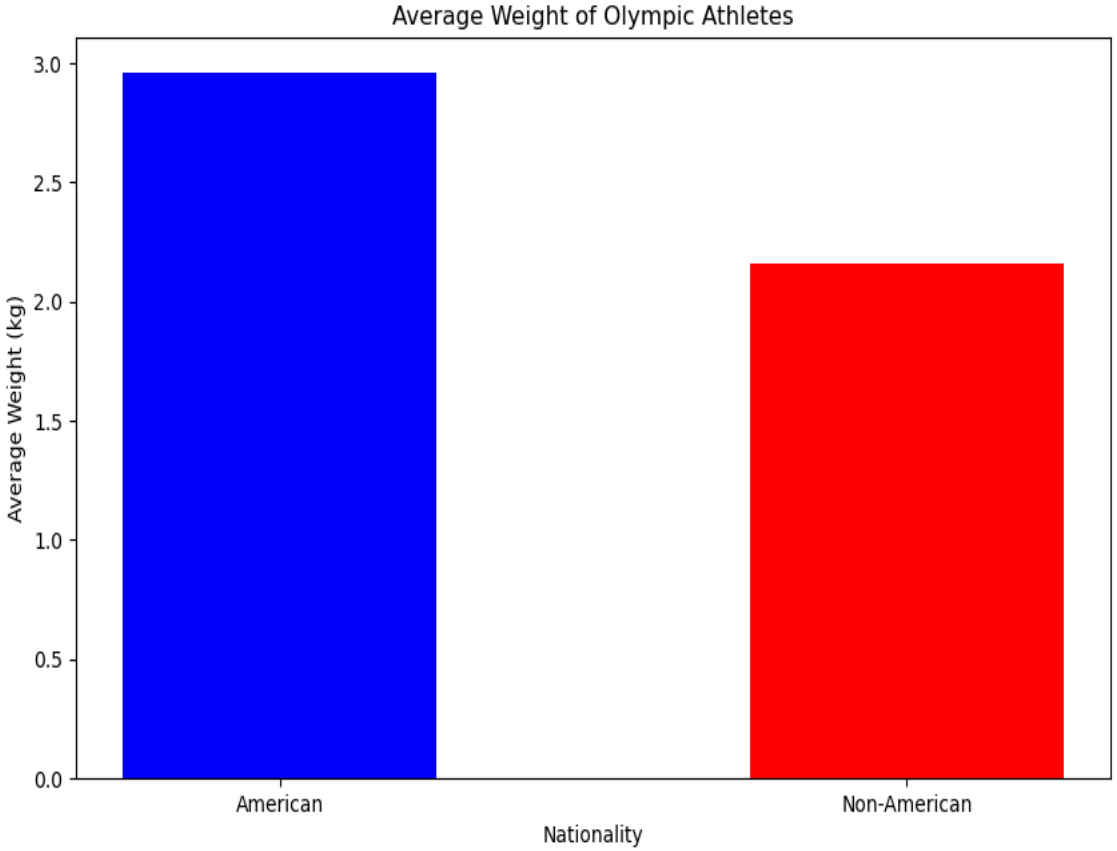
Insights: Gender

- Overall gender ratio: 51% male, 49% female
  - Suggests gender parity among athletes

- Greater gender disparity among low-GDP countries (60% male)
  - Most common sports in these countries are male-dominated
  - Gender norms in different cultures limit women’s ability to participate in certain sports

Athlete Demographics: Height and Weight

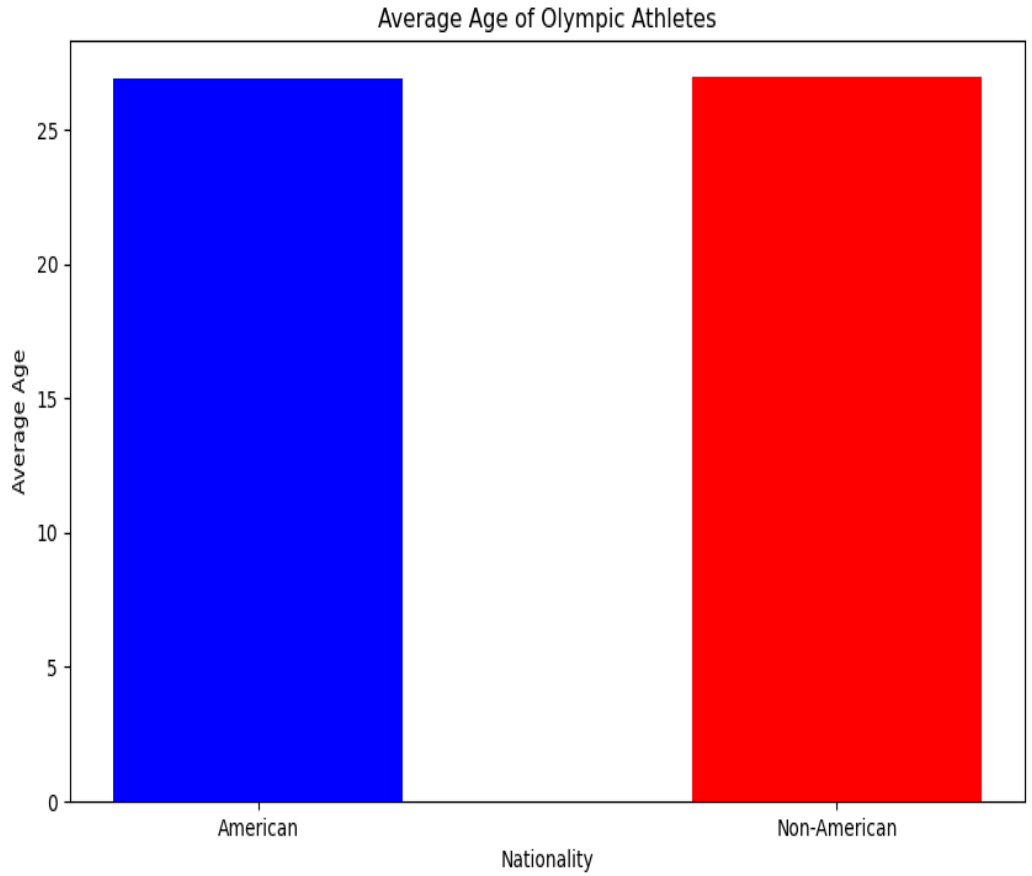


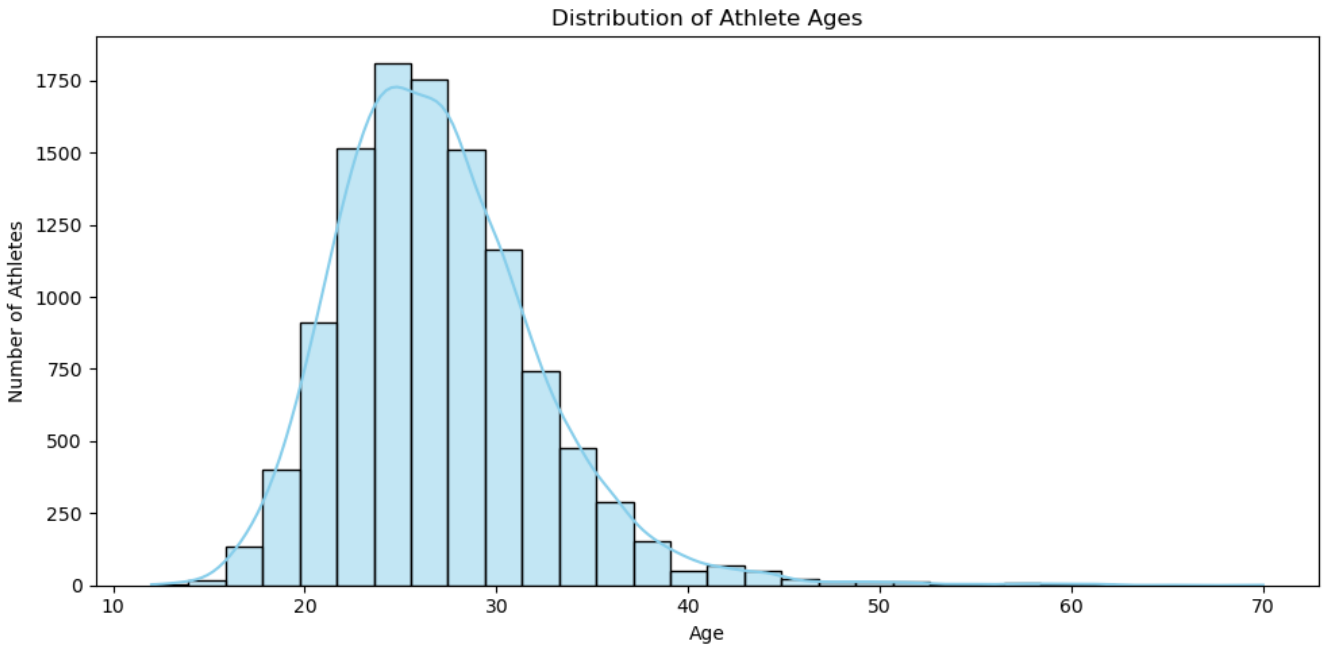


Insights: Height and Weight

- American athletes have greater average height and weight compared to non-American athletes
  - Greater resources to invest in factors that promote growth and development (training, nutrition, etc.)
  - Genetically diverse population contributes to greater height and weight
  - American athletes highly represented in sports that favor more height (e.g., Basketball) or weight (e.g., Wrestling).

Athlete Demographics: Age

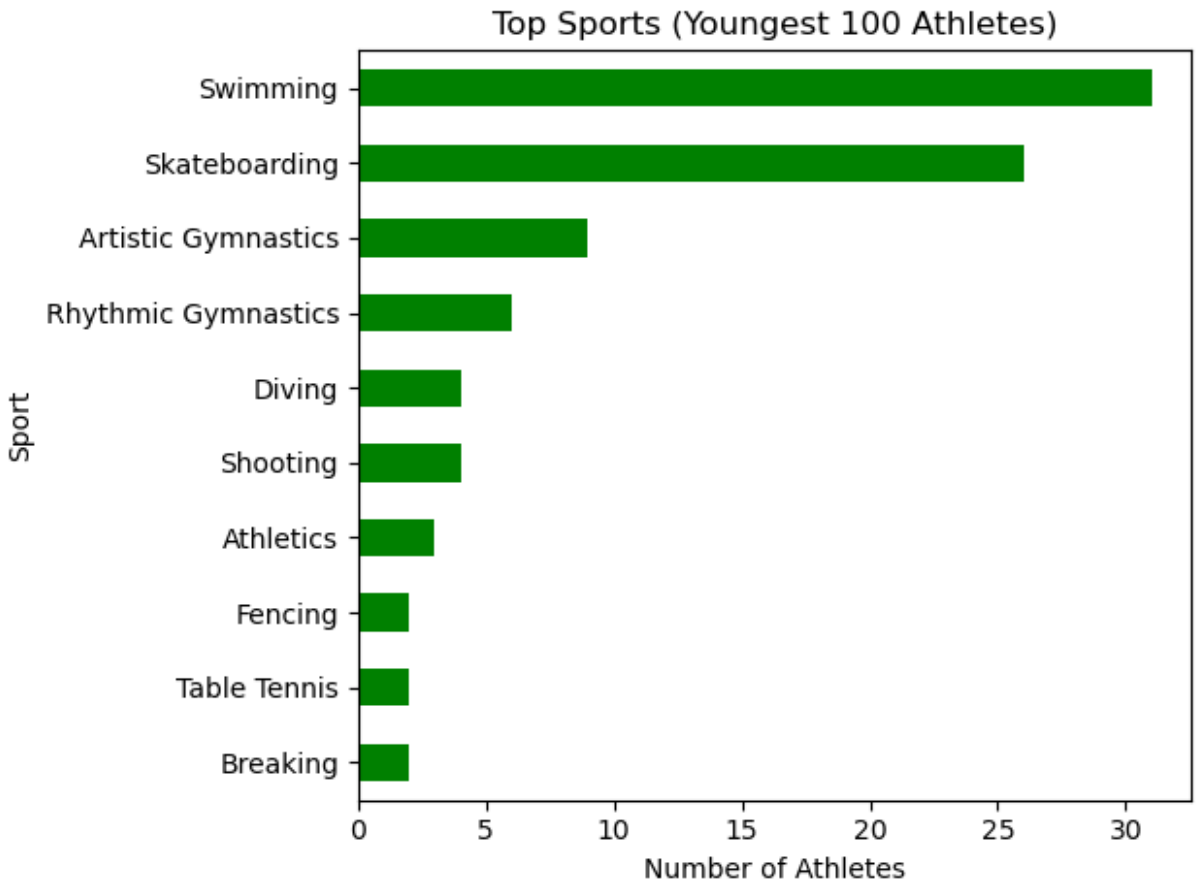


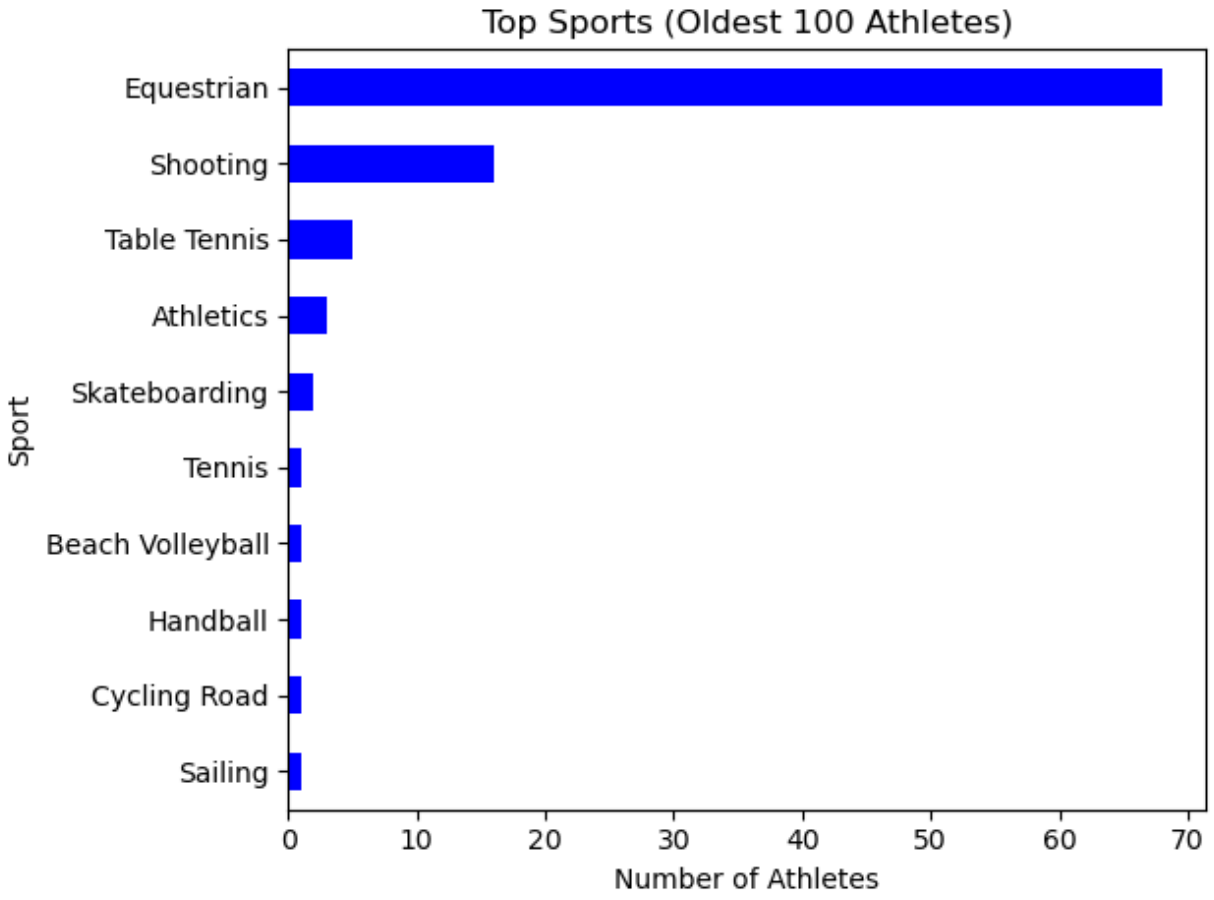


Insights: Age

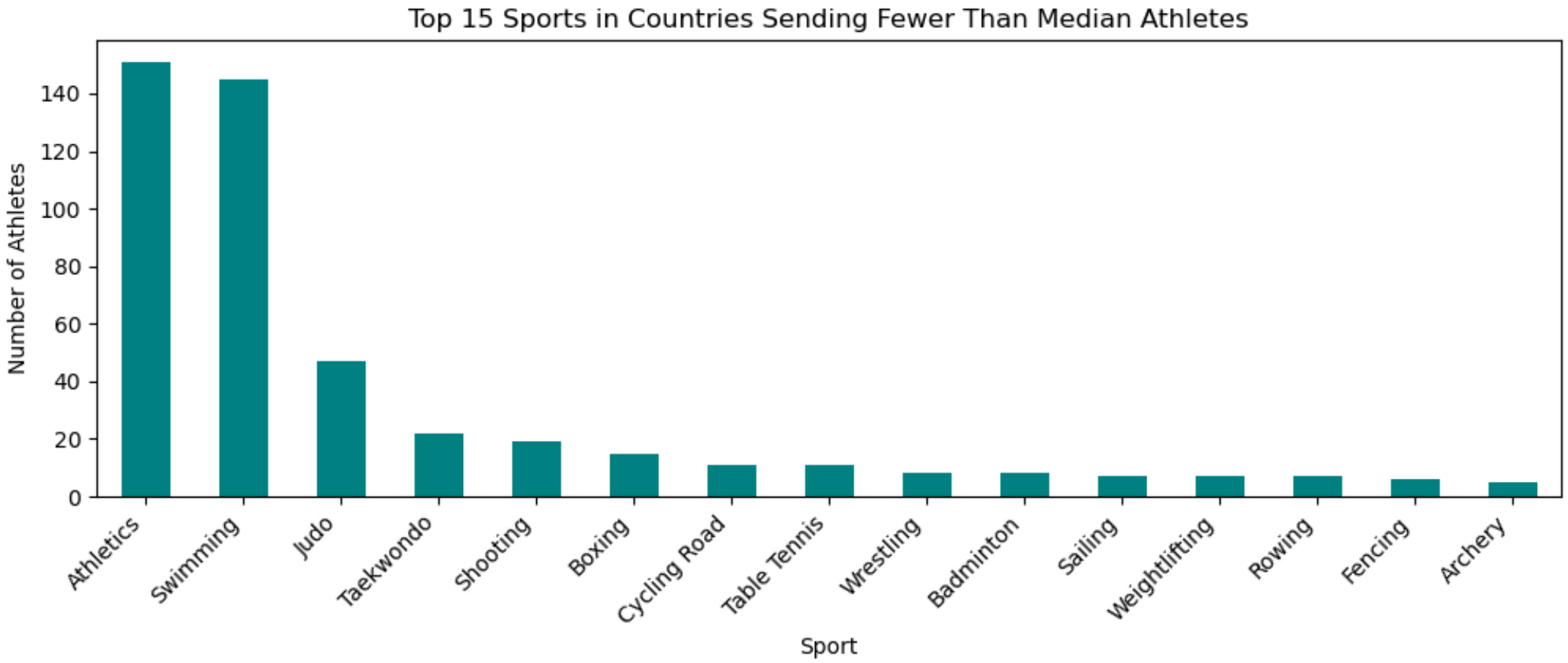
- Little to no difference in average age among American vs. non-American athletes.
- Majority of athletes fall in 20-30 age range.
- Younger athletes yield highest performance and are most likely to win medals overall.

Sport Participation: Age and GDP

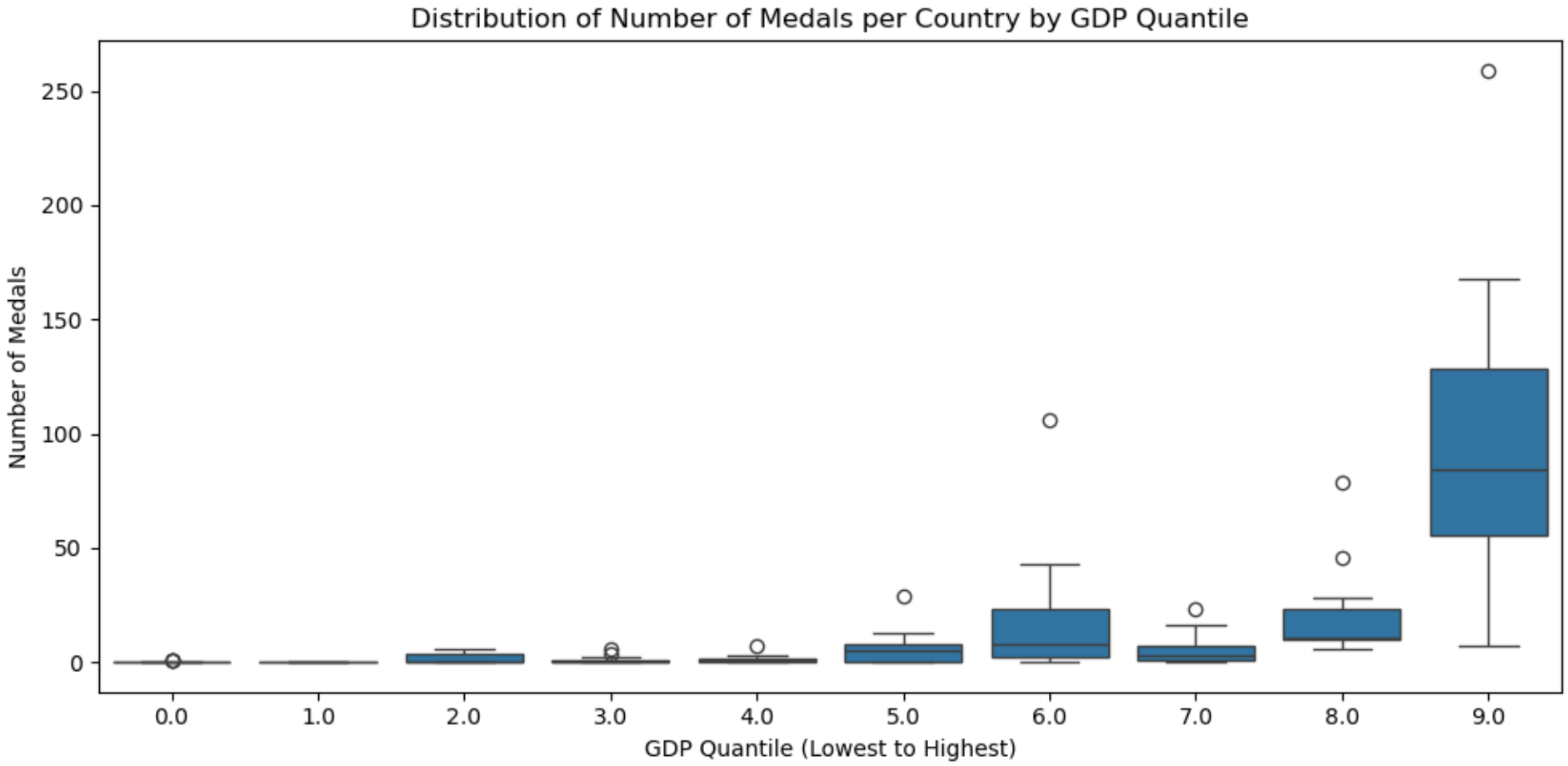




Sport Participation (Below Median GDP)



Sport Participation (GDP) cont.



Quantile 9: Highest GDP Countries

Sport Participation Insights (Age and GDP)

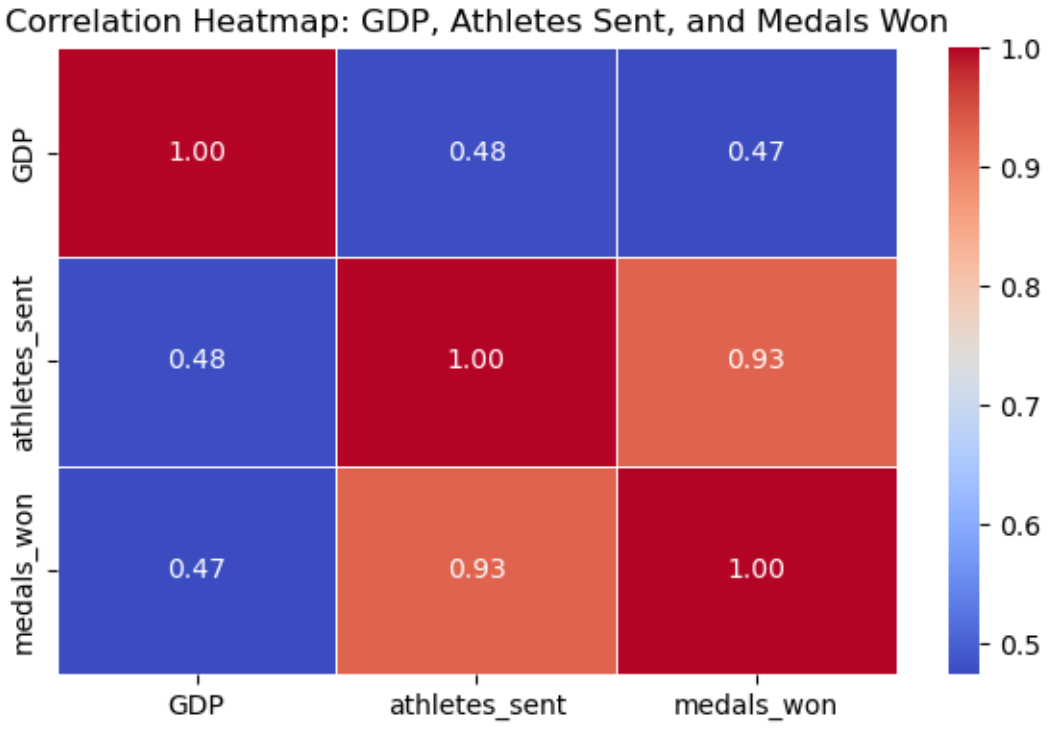
- Among lowest-GDP countries, top sports are Swimming and Athletics
  - Low barrier to entry and cheaper infrastructure
  - Highly-rooted in education in most countries
  - Lower-income countries often targeted by international bodies to promote participation in these sports

- Most common sport for younger athletes: Swimming
  - Peak performance age is in 20’s, with some athletes in their teens
- Most common sport for older athletes: Equestrian
  - Favors long-term skills and experience over physicality

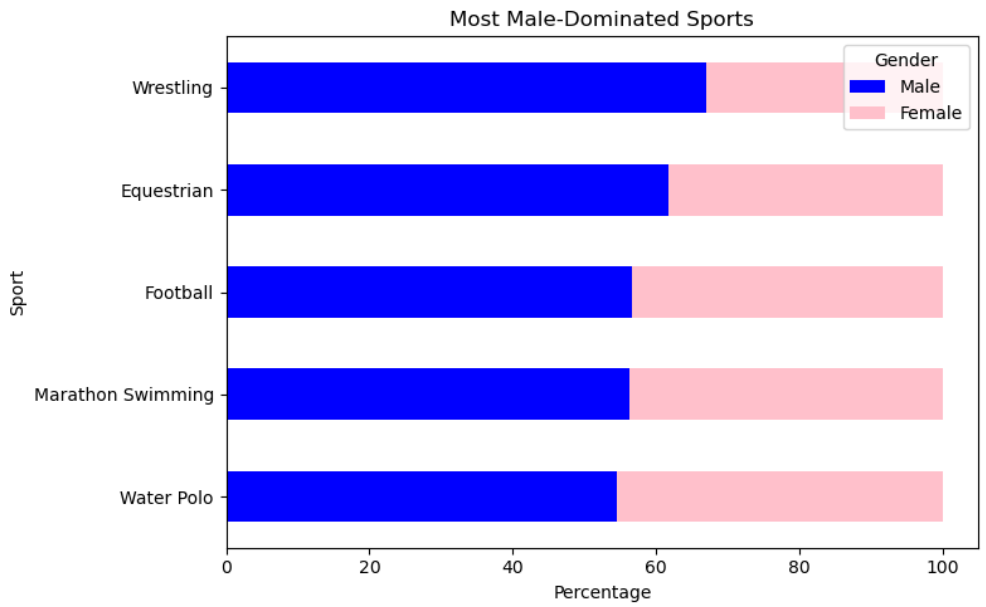
Sport Participation Insights (Age and GDP) cont.

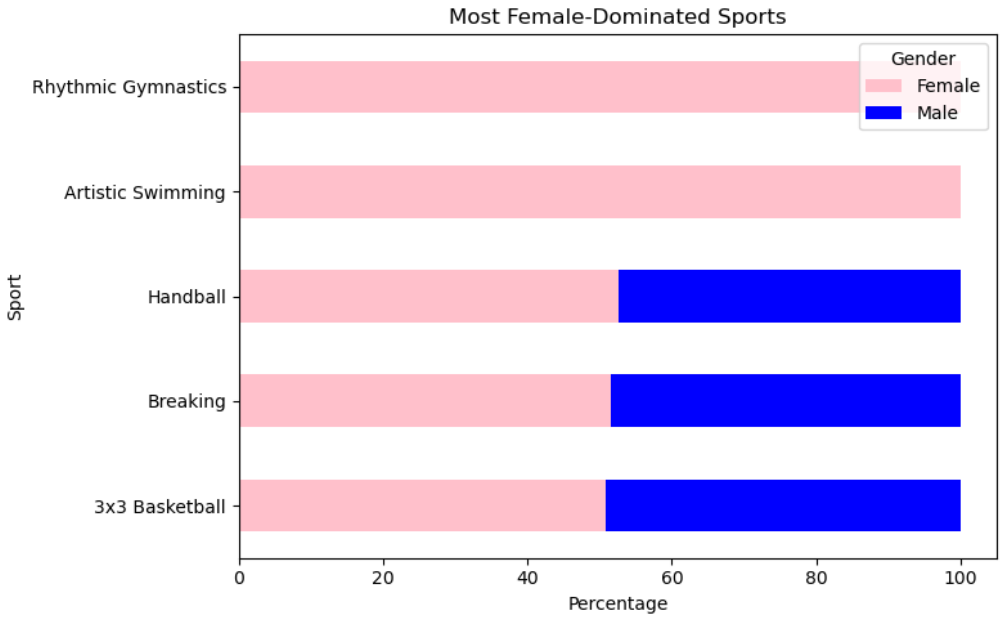
- Moderate correlations between GDP and athletes sent to Olympics (0.48) and medals won (0.47).

- High correlation between number of athletes sent and medals won (0.93).
  - Suggests higher sheer number of athletes leads to more medals won
  - However, other factors are in play:
    - Strategic investments in sports that are most likely to yield medals



Sport Participation (Gender)





Sport Participation Insights (Gender)

- Most male-dominated sports: Wrestling and Equestrian
- Most female-dominated sports: Rhythmic Gymnastics and Artistic Swimming
- Presence of gender bias and social norms (for both genders)
- Disparity in socioeconomic factors that favor men (e.g., Equestrian)

Statistical Analyses

- Age
  - There is no significant difference in ages of medallists vs. non-medallists ($p = 0.24$)
- Height
  - At least one sport has a significant difference in athlete height compared to other sports ($p < 0.01$)
- Gender
  - There is no significant difference in medal-winning rates between male and female athletes ($p = 0.50$)
- GDP
  - There is a significant relationship between a country's GDP level and medal count ($p < 0.01$)

Statistical Analysis Insights

- Most athletes come from similar age range, reducing the variance in the sample.
- Medal-winning ability likely based more on skill.
- Performance in some sports favor athletes of shorter or taller heights, depending on demands.
- Gender parity creates more equal representation of male and female athletes.
- Higher GDP leads to more resources for higher-cost sports and higher athlete representation.

Conclusions

- GDP is strongest predictor of success in Olympics
  - More resources to invest in athletes, training, nutrition, and sports infrastructure
  - Contributes to physical characteristics (e.g., height and weight) that can improve performance in certain sports
- Gender parity is present overall
  - However, some disparity still exists within specific sports and countries
- Some sports favor one gender over the other (physical characteristics and/or societal/cultural norms)

Recommendations

- International Olympic Committee/Federations should continue to prioritize gender-equal opportunities for funding
  - Focus on countries with greater than 60% male athlete representation and/or low GDP
- Invest more in low-resource countries to focus on sports that can lead to higher numbers of medals
  - Prioritize improving access to state of the art training facilities and programs
- Explore methods to address cultural difference that may prohibit athletes from participating in certain sports
  - E.g., standardized uniforms